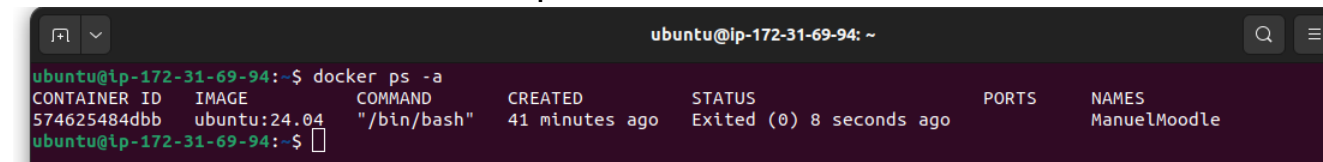


MI IP 192.168.129.210/24

Ejercicio 5: Instalación del gestor de contenidos Moodle en contenedor Docker (2 puntos)

Creamos el contenedor

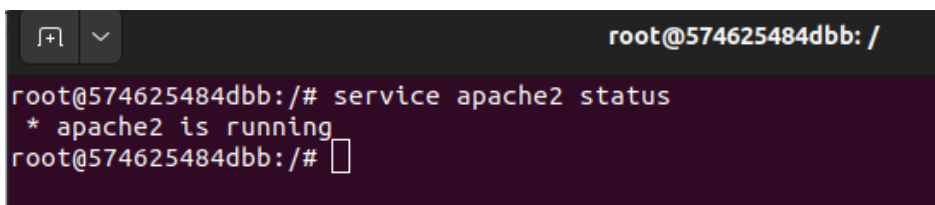
```
docker run -it --name ManuelMoodle -p 80:80 ubuntu:24.04 /bin/bash
```



```
ubuntu@ip-172-31-69-94: ~  
ubuntu@ip-172-31-69-94:~$ docker ps -a  
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS      NAMES  
574625484dbb   ubuntu:24.04  "/bin/bash"             41 minutes ago  Exited (0)    8 seconds ago  ManuelMoodle  
ubuntu@ip-172-31-69-94:~$
```

Instalar dependencias iniciales

```
apt update  
apt install vim  
apt install wget  
apt install unzip
```



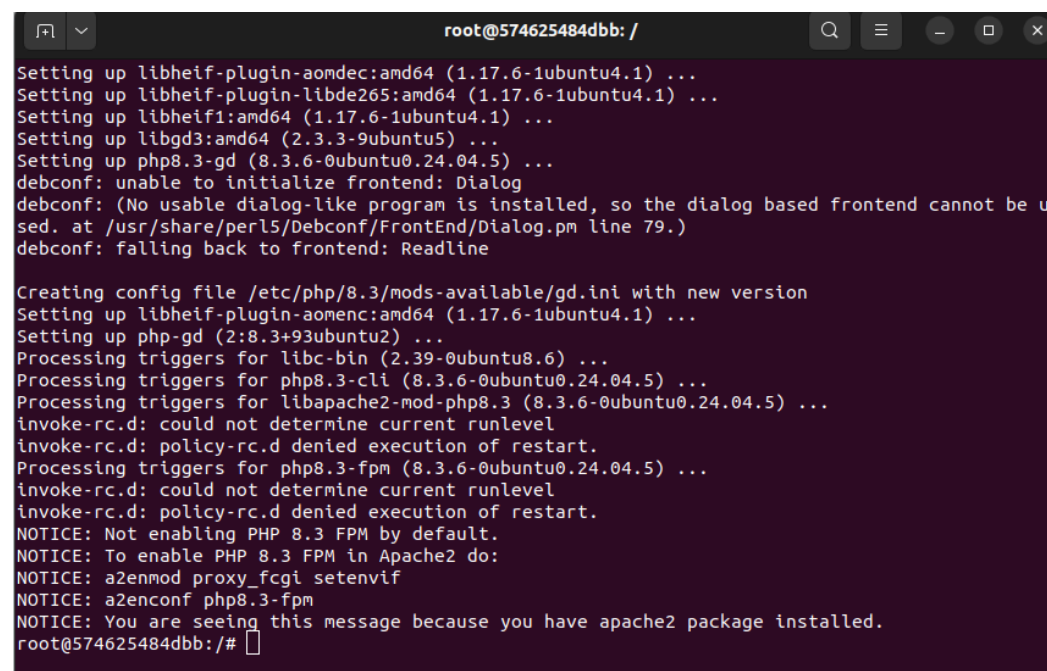
```
root@574625484dbb: /  
root@574625484dbb:/# service apache2 status  
* apache2 is running  
root@574625484dbb:/#
```

Instalar Apache 2

```
apt install apache2  
service apache2 start
```

Instalar PHP

```
apt install php php-cli php-fpm php-mysql php-xml php-xmlrpc \  
php-gd php-curl php-zip php-intl php-mbstring php-soap php-bcmath \  
php-ldap php-readline php-tokenizer php-common php-opcache libapache2-mod-php -y
```



```
root@574625484dbb: /  
Setting up libheif-plugin-aomdec:amd64 (1.17.6-1ubuntu4.1) ...  
Setting up libheif-plugin-libde265:amd64 (1.17.6-1ubuntu4.1) ...  
Setting up libheif1:amd64 (1.17.6-1ubuntu4.1) ...  
Setting up libgd3:amd64 (2.3.3-9ubuntu5) ...  
Setting up php8.3-gd (8.3.6-0ubuntu0.24.04.5) ...  
debconf: unable to initialize frontend: Dialog  
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 79.)  
debconf: falling back to frontend: Readline  
  
Creating config file /etc/php/8.3/mods-available/gd.ini with new version  
Setting up libheif-plugin-aomenc:amd64 (1.17.6-1ubuntu4.1) ...  
Setting up php-gd (2:8.3+93ubuntu2) ...  
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...  
Processing triggers for php8.3-cli (8.3.6-0ubuntu0.24.04.5) ...  
Processing triggers for libapache2-mod-php8.3 (8.3.6-0ubuntu0.24.04.5) ...  
invoke-rc.d: could not determine current runlevel  
invoke-rc.d: policy-rc.d denied execution of restart.  
Processing triggers for php8.3-fpm (8.3.6-0ubuntu0.24.04.5) ...  
invoke-rc.d: could not determine current runlevel  
invoke-rc.d: policy-rc.d denied execution of restart.  
NOTICE: Not enabling PHP 8.3 FPM by default.  
NOTICE: To enable PHP 8.3 FPM in Apache2 do:  
NOTICE: a2enmod proxy_fcgi setenvif  
NOTICE: a2enconf php8.3-fpm  
NOTICE: You are seeing this message because you have apache2 package installed.  
root@574625484dbb:/#
```

Instalar MariaDB y Crear base de datos para Moodle

```
apt install mariadb-server
service mariadb status
service mariadb start
mariadb
```

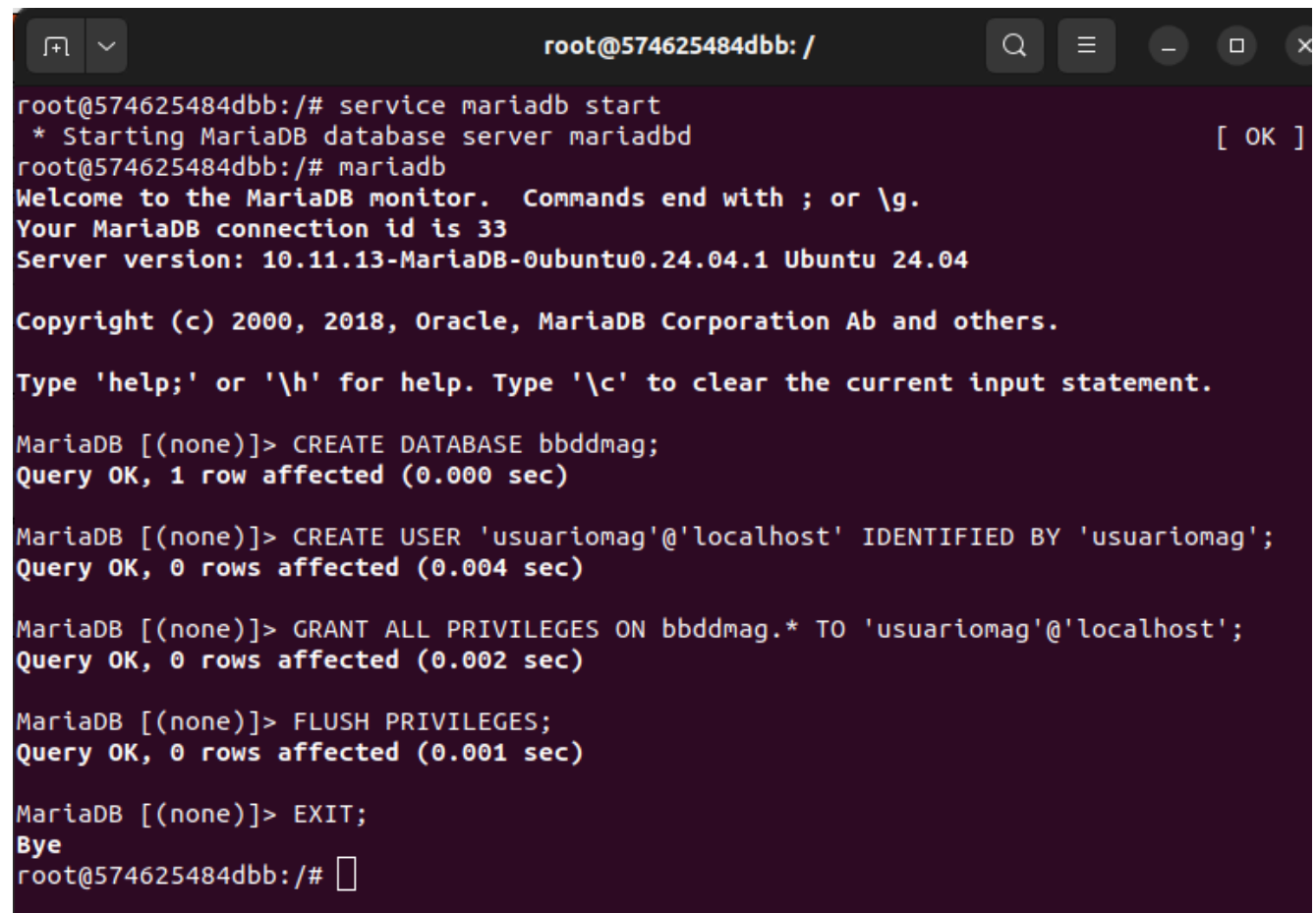
```
CREATE DATABASE bbddmag;
```

```
CREATE USER 'usuariomag'@'localhost' IDENTIFIED BY 'usuariomag';
```

```
GRANT ALL PRIVILEGES ON bbddmag.* TO 'usuariomag'@'localhost';
```

```
FLUSH PRIVILEGES;
```

```
EXIT;
```

A screenshot of a terminal window with a dark background and light-colored text. The window title bar shows 'root@574625484dbb: /' and standard window controls. The terminal output shows the following sequence of commands and responses:
1. `root@574625484dbb:/# service mariadb start`
 * Starting MariaDB database server mariadbd [OK]
2. `root@574625484dbb:/# mariadb`
 Welcome to the MariaDB monitor. Commands end with ; or \g.
 Your MariaDB connection id is 33
 Server version: 10.11.13-MariaDB-0ubuntu0.24.04.1 Ubuntu 24.04

 Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

 Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

 MariaDB [(none)]> CREATE DATABASE bbddmag;
 Query OK, 1 row affected (0.000 sec)

 MariaDB [(none)]> CREATE USER 'usuariomag'@'localhost' IDENTIFIED BY 'usuariomag';
 Query OK, 0 rows affected (0.004 sec)

 MariaDB [(none)]> GRANT ALL PRIVILEGES ON bbddmag.* TO 'usuariomag'@'localhost';
 Query OK, 0 rows affected (0.002 sec)

 MariaDB [(none)]> FLUSH PRIVILEGES;
 Query OK, 0 rows affected (0.001 sec)

 MariaDB [(none)]> EXIT;
 Bye
 root@574625484dbb:/#

Instalar Moodle

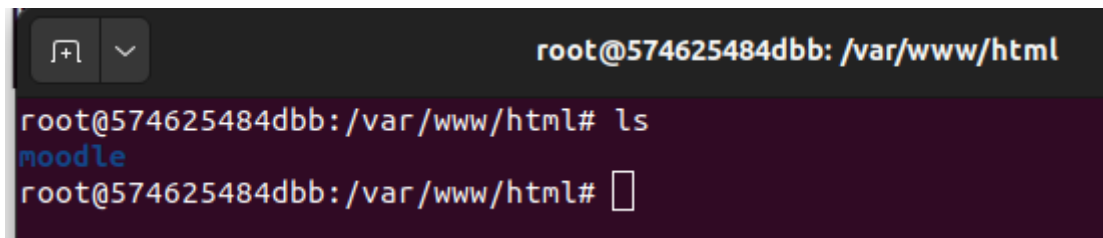
```
cd /var/www/html
```

```
wget https://download.moodle.org/download.php/direct/stable501/moodle-latest-5
```

```
unzip moodle-latest-501.zip
```

```
rm moodle-latest-501.zip
```

```
rm index.html
```

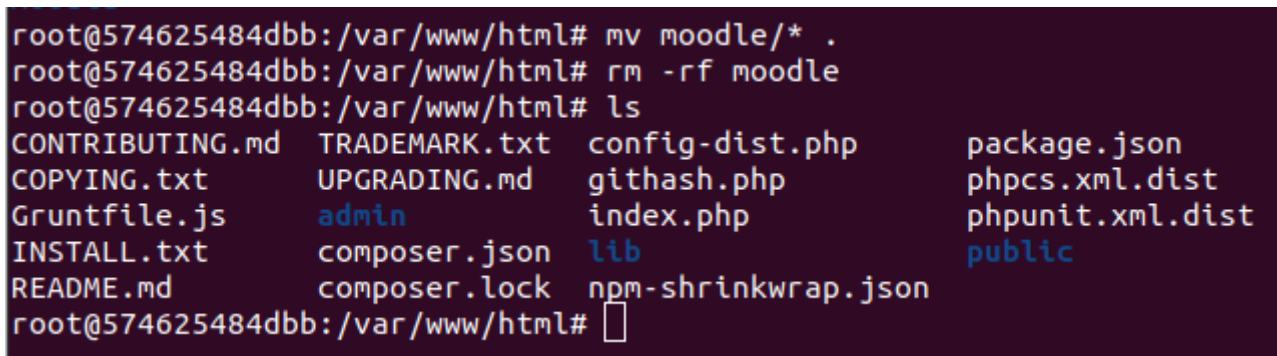


```
root@574625484dbb: /var/www/html
root@574625484dbb:/var/www/html# ls
moodle
root@574625484dbb:/var/www/html#
```

Mover los archivos

```
mv moodle/* .
```

```
rm -rf moodle
```



```
root@574625484dbb:/var/www/html# mv moodle/* .
root@574625484dbb:/var/www/html# rm -rf moodle
root@574625484dbb:/var/www/html# ls
CONTRIBUTING.md  TRADEMARK.txt  config-dist.php  package.json
COPYING.txt       UPGRADING.md   githash.php      phpcs.xml.dist
Gruntfile.js      admin          index.php        phpunit.xml.dist
INSTALL.txt       composer.json  lib              public
README.md         composer.lock  npm-shrinkwrap.json
root@574625484dbb:/var/www/html#
```

Configuración de Moodle

```
cp config-dist.php config.php
```

```
vim config.php
```

```
$CFG->dbtype = 'mariadb';
```

```
$CFG->dblibrary = 'native';
```

```
$CFG->dbhost = 'localhost';
```

```
$CFG->dbname = 'bbddmag';
```

```
$CFG->dbuser = 'usuariomag';
```

```
$CFG->dbpass = 'usuariomag';
```

```
$CFG->wwwroot = 'http://98.82.127.52'; <-- IP de la Instancia
```

```
$CFG->dataroot = '/var/moodledata';
```

```

root@574625484dbb: /var/www/html

//=====
// 1. DATABASE SETUP
//=====
// First, you need to configure the database where all Moodle data      //
// will be stored. This database must already have been created        //
// and a username/password created to access it.                        //

$CFG->dbtype      = 'mysqli';           // 'pgsql', 'mysqli', 'auroramySQL', or 'sqlsrv'
$CFG->dblibrary    = 'native';          // 'native' only at the moment
$CFG->dbhost       = 'localhost';        // eg 'localhost' or 'db.isp.com' or IP
$CFG->dbname       = 'bbddmag';          // database name, eg moodle
$CFG->dbuser       = 'usuariomag';       // your database username
$CFG->dbpass       = 'usuariomag';       // your database password
$CFG->prefix       = 'mdl_';             // prefix to use for all table names
$CFG->dboptions    = [
    'dbpersist' => false,               // Should persistent database connections be

```

```
$CFG->wwwroot      = 'http://98.82.127.52';
```

```
$CFG->dataroot     = '/var/moodledata';
```

Configuración de PHP

vim /etc/php/8.3/apache2/php.ini

buscamos:

```
; max_input_vars = 1000
```

y dejalo asi:

```
max_input_vars = 5000
```

```

; How many GET/POST/COOKIE input variables may be accepted
max_input_vars = 5000

```

Dar Permisos

```
chown -R www-data:www-data /var/www/html
```

```
chmod -R 755 /var/www/html
```

```
mkdir /var/moodledata
```

```
chown -R www-data:www-data /var/moodledata
```

```
chmod -R 755 /var/moodledata
```

```

root@574625484dbb: /var/www/html

root@574625484dbb:/var/www/html# chown -R www-data:www-data /var/www/html
root@574625484dbb:/var/www/html# chmod -R 755 /var/www/html
root@574625484dbb:/var/www/html# mkdir /var/moodledata
root@574625484dbb:/var/www/html# chown -R www-data:www-data /var/moodledata
root@574625484dbb:/var/www/html# chmod -R 755 /var/moodledata
root@574625484dbb:/var/www/html#

```

Configurar Apache

```
a2ensite 000-default.conf
a2enmod rewrite
service apache2 restart
vim /etc/apache2/sites-available/000-default.conf
Escribe:
<Directory /var/www/html>
AllowOverride All
Require all granted
</Directory>
y añade:
```

```
DocumentRoot /var/www/html/public
service apache2 restart
```

```
root@574625484dbb: /var/www/html

root@574625484dbb:/var/www/html# a2ensite 000-default.conf
Site 000-default already enabled
root@574625484dbb:/var/www/html# a2enmod rewrite
Enabling module rewrite.
To activate the new configuration, you need to run:
    service apache2 restart
root@574625484dbb:/var/www/html# service apache2 restart
 * Restarting Apache httpd web server apache2
AH00558: apache2: Could not reliably determine the server's fully qualified domain name,
please add the 'ServerName' directive globally to suppress this message

root@574625484dbb:/var/www/html#
```

```
root@574625484dbb: /var/www/html

<VirtualHost *:80>
    # The ServerName directive sets the request scheme, hostname and port that
    # the server uses to identify itself. This is used when creating
    # redirection URLs. In the context of virtual hosts, the ServerName
    # specifies what hostname must appear in the request's Host: header to
    # match this virtual host. For the default virtual host (this file) this
    # value is not decisive as it is used as a last resort host regardless.
    # However, you must set it for any further virtual host explicitly.
    #ServerName www.example.com

    ServerAdmin webmaster@localhost
    DocumentRoot /var/www/html/public

    # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
    # error, crit, alert, emerg.
    # It is also possible to configure the loglevel for particular
    # modules, e.g.
    #LogLevel info ssl:warn

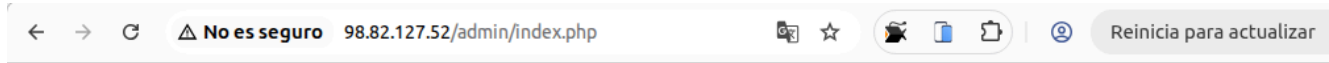
    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined

    # For most configuration files from conf-available/, which are
    # enabled or disabled at a global level, it is possible to
    # include a line for only one particular virtual host. For example the
    # following line enables the CGI configuration for this host only
    # after it has been globally disabled with "a2disconf".
    #Include conf-available/serve-cgi-bin.conf

    <Directory /var/www/html>
        AllowOverride All
        Require all granted
    </Directory>
</VirtualHost>
```

```
root@574625484dbb: /var/www/html
root@574625484dbb:/var/www/html# vim /etc/apache2/sites-available/000-default.conf
root@574625484dbb:/var/www/html# service apache2 restart
* Restarting Apache httpd web server apache2
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.17.0.2. Set the 'ServerName' directive globally to suppress this message
[ OK ]
root@574625484dbb:/var/www/html#
```

http://98.82.127.52 <-- IP de la Instancia



Installation

Moodle - Modular Object-Oriented Dynamic Learning Environment

Copyright notice

Copyright (C) 1999 onwards Martin Dougiamas (<https://moodle.com>)

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Confirm

Have you read these conditions and understood them?

Cancel

Continue

Installation - Moodle 5.1+ (Build: 20251205)

Moodle 5.1+ (Build: 20251205)

For information about this version of Moodle, please see the online [Release Notes](#)

Server checks

Name	Information	Report	Plugin	Status
unicode		 must be installed and enabled ↗		
database	mariadb (10.11.13-MariaDB-0ubuntu0.24.04.1)	 version 10.11.0 is required and you are running 10.11.13 ↗		
php		 version 8.2.0 is required and you are running 8.3.6 ↗		
pcreunicode		 should be installed and enabled for best results ↗		
php_extension	iconv	 must be installed and enabled ↗		
php_extension	mbstring	 must be installed and enabled ↗		
php_extension	curl	 must be installed and enabled ↗		
php_extension	openssl	 must be installed and enabled ↗		
php_extension	tokenizer	 should be installed and enabled for best results ↗		
php_extension	soap	 should be installed and enabled for best results ↗		
php_extension	ctype	 must be installed and enabled ↗		
php_extension	zip	 must be installed and enabled ↗		
php_extension	zlib	 must be installed and enabled ↗		
php_extension	gd	 must be installed and enabled ↗		
php_extension	simplexml	 must be installed and enabled ↗		
php_extension	enl	 must be installed and enabled ↗		

Installation

System

Success (16.02 seconds)

aiplacement_courseassist

Success (0.07 seconds)

aiplacement_editor

Success (0.06 seconds)

aiprovider_azureai

Success (0.03 seconds)

aiprovider_deepseek

Success (0.03 seconds)

aiprovider_ollama

Success (0.03 seconds)

Installation

On this page you should configure your main administrator account which will have complete control over the site. Make sure you give it a secure username and password as well as a valid email address. You can create more admin accounts later on.

[Expand all](#)

▼ General

Username ⓘ

Choose an authentication method ⓘ Manual accounts

The password must have at least 8 characters, at least 1 digit(s), at least 1 lower case letter(s), at least 1 upper case letter(s), at least 1 special character(s) such as *, -, or #

New password ⓘ ⓘ ⓘ

Required

☐ Force password change ⓘ

First name ⓘ

Last name ⓘ

Email address ⓘ

Email visibility ⓘ ▼

City/town

Select a country ▼

Manuel Examen

Search 🔍

- General
- Users
- Courses
- Grades
- Plugins
- Appearance
- Server
- Reports
- Development

Complete your site registration

Registering your site involves periodically sending [anonymised site stats](#) to Moodle HQ. This helps us gain insights into how you're using Moodle, and make decisions about new features and plugins.

Plus, it enables you to:

- Stay up to date with important updates, like security alerts and new available releases.
- Activate mobile push notifications for the [Moodle App](#).
- Increase the visibility of your site by featuring it on the list of registered Moodle sites in your country (optional).

- [🔗 View global Moodle stats](#)
- [📍 Discover other sites in my country](#)

Expand all

▾

Registration information

Organisation name

?

Manuel Examen

Organisation type

?

Do not share ▾

Site listing

?

Do not list my site ▾

Description

?

Language

?

English ▾

Dashboard

Welcome, Manuel! 🙌

Timeline

Next 7 days ▼

Sort by dates ▼

Search by activity type or name



No in-progress courses

Calendar

All courses ▼

New event

◀ November

December 2025

January ▶

Mon	Tue	Wed	Thu	Fri	Sat	Sun
1	2	3	4	5	6	7
8	9	10	11	12	13	14

Ejercicio 6: Instalación de magento en contenedor Docker 2 puntos)

Creamos el contenedor

```
docker run -it --name ManuelMagento -p 8080:80 ubuntu:24.04 /bin/bash
```

Instalar dependencias iniciales

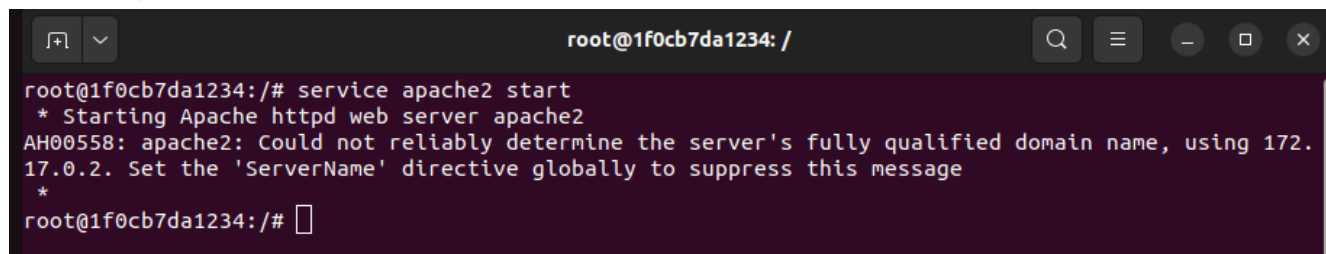
```
apt update
apt install vim
apt install wget
apt install unzip
```

Añadir el repositorio de PHP 5.6

```
apt install -y software-properties-common
add-apt-repository ppa:ondrej/php -y
apt update
```

Instalar Apache 2

```
apt install apache2
service apache2 start
```

A terminal window with a dark background. The title bar shows 'root@1f0cb7da1234: /'. The terminal content shows the command 'service apache2 start' being executed. The output indicates that Apache is starting successfully, with a warning message from AH00558 about the server's fully qualified domain name. The prompt returns to 'root@1f0cb7da1234: /#'.

```
root@1f0cb7da1234:/# service apache2 start
* Starting Apache httpd web server apache2
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.17.0.2. Set the 'ServerName' directive globally to suppress this message
*
root@1f0cb7da1234:/#
```

Instalar PHP 5.6 + extensiones necesarias para Magento

```
apt install -y php5.6 php5.6-common php5.6-cli php5.6-fpm \
php5.6-mysql php5.6-curl php5.6-xml php5.6-intl php5.6-gd \
php5.6-mbstring php5.6-zip libapache2-mod-php5.6 \
php5.6-mcrypt php5.6-soap
```

Instalar MariaDB y Crear base de datos para Magento

```
apt install mariadb-server
service mariadb status
service mariadb start
mariadb
```

```
CREATE DATABASE bbddmag;
```

```
CREATE USER 'usuariomag'@'localhost' IDENTIFIED BY 'usuariomag';
```

```
GRANT ALL PRIVILEGES ON bbddmag.* TO 'usuariomag'@'localhost';
```

```
FLUSH PRIVILEGES;
```

```
EXIT;
```

```
root@1f0cb7da1234: /

root@1f0cb7da1234:/# mariadb
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 33
Server version: 10.11.13-MariaDB-0ubuntu0.24.04.1 Ubuntu 24.04

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE bbddmag;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> CREATE USER 'usuariomag'@'localhost' IDENTIFIED BY 'usuariomag';
Query OK, 0 rows affected (0.002 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON bbddmag.* TO 'usuariomag'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> EXIT;
Bye
root@1f0cb7da1234:/#
```

Instalar Magento

```
cd /var/www/html
rm index.html
wget https://github.com/magento2-download/magento-CE-1.9.x/releases/download/1.9.3.8/magento-1.9.3.8.zip
unzip magento-1.9.3.8.zip
rm magento-1.9.3.8.zip
```

```
root@1f0cb7da1234: /var/www/html

root@1f0cb7da1234:/var/www/html# rm magento-1.9.3.8.zip
root@1f0cb7da1234:/var/www/html# ls
magento
root@1f0cb7da1234:/var/www/html#
```

Mover los archivos

```
mv magento/* .
mv magento/. * . 2>/dev/null || true
rm -rf magento
```

```
root@1f0cb7da1234:/var/www/html# mv magento/* .
root@1f0cb7da1234:/var/www/html# mv magento/. * . 2>/dev/null || true
root@1f0cb7da1234:/var/www/html# rm -rf magento
root@1f0cb7da1234:/var/www/html# ls
LICENSE.html      api.php    dev        get.php      install.php  media        var
LICENSE.txt       app        downloader includes      js           php.ini.sample
LICENSE_AFL.txt   cron.php   errors     index.php    lib          shell
RELEASE_NOTES.txt cron.sh    favicon.ico index.php.sample mage         skin
root@1f0cb7da1234:/var/www/html#
```

Dar Permisos

```
find . -type d -exec chmod 755 {} \;  
find . -type f -exec chmod 644 {} \;  
chmod -R 755 var media includes  
chown -R www-data:www-data /var/www/html
```

```
root@1f0cb7da1234:/var/www/html# find . -type d -exec chmod 755 {} \;  
root@1f0cb7da1234:/var/www/html# find . -type f -exec chmod 644 {} \;  
root@1f0cb7da1234:/var/www/html# chmod -R 755 var media includes  
root@1f0cb7da1234:/var/www/html# chown -R www-data:www-data /var/www/html  
root@1f0cb7da1234:/var/www/html#
```

Configurar Apache

```
a2enmod rewrite  
service apache2 restart  
vim /etc/apache2/sites-available/000-default.conf  
Escribe:  
<Directory /var/www/html>  
AllowOverride All  
Require all granted  
</Directory>  
service apache2 restart
```

```
<Directory /var/www/html>  
    AllowOverride All  
    Require all granted  
</Directory>
```

sgr-08a431aef3713bdce	HTTP	TCP	80	Persona...	Q		Eliminar
-	TCP personalizado	TCP	8080	Anywh...	Q 0.0.0.0/0		Eliminar
					Q 0.0.0.0/0		Eliminar

← → ↻ ⚠ No es seguro 98.82.127.52:8080/index.php/install/ ⓘ ⚙ ⭐ 🖨 📄 🔄 ⌛ Reinicia para actualizar



Installation

Download
License Agreement
Localization
Configuration
Create Admin Account
You're All Set!

Having trouble installing Magento?
Check out our [Installation Guide](#)

Welcome to Magento's Installation Wizard!

Open Software License ("OSL") v. 3.0

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- to distribute or communicate copies of the Original Work and Derivative Works to the public, with the proviso that copies of Original Work or Derivative Works that You distribute or communicate shall be licensed under this Open Software License

☐ I agree to the above terms and conditions.

Continue

Help Us to Keep Magento Healthy - Report All Bugs (ver. 1.9.3.8)

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Installation

Download
License Agreement
Localization
Configuration
Create Admin Account
You're All Set!

Having trouble installing Magento?
Check out our [Installation Guide](#)

Configuration

Database Connection

Database Type

MySQL

Host *

localhost

You can specify server port, ex.: localhost:3307
If you are not using default UNIX socket, you can specify
it here instead of host, ex.: /var/run/mysqld/mysqld.sock

Database Name *

bbddmag

User Name *

usuariomag

User Password

Tables Prefix

(Optional. Leave blank for no prefix)

Web access options

Base URL *

http://98.82.127.52:8080/

Admin Path *

admin

Additional path added after Base URL to access your Administrative Panel (e.g. admin, backend, control etc.).

☒ Enable Charts

Enable this option if you want the charts to be displayed on Dashboard.

☒ Skip Base URL Validation Before the Next Step

Check this box only if it is not possible to automatically validate the Base URL.

☐ Use Web Server (Apache) Rewrites

You could enable this option to use web server rewrites functionality for improved search engines optimization.
Please make sure that `mod_rewrite` is enabled in Apache configuration.

☐ Use Secure URLs (SSL)

Enable this option only if you have SSL available.

Session Storage Options